

How Machines Pronounce Korean Correctly

Context & Semantics-Aware Real-time Text Preprocessing Engine v2.0 (C++ Native & Lightweight Neural Backbone)

ARCHITECTURE

C++ Native Core

NEURAL BACKBONE

miniBERT KD/INT8

INFERENCE LATENCY

1.86 ms (Single CPU)

HARDWARE OVERHEAD

0 MB GPU VRAM

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Slide 02. Representative Demo & Context Disambiguation

Real-time disambiguation and phonetic transcription of complex contexts, numeric identifiers, and out-of-vocabulary terms

1. Input Natural Text // input_raw_text

2026.8.25 오후 3:10에 2호선 3번 출구에 있는 ABCTechnology라는 회사에서 Galaxy Z fold8 ultra를 샀어요.

// Generated Phonemic Target Stream (target_phoneme_stream)

이쳐니심능년 파월 이시보일 오후 세 시 십 부네 이호선 삼번 출구에 인는 에이비씨 테크놀로지라는 회사에서 갤럭시 지 폴드에인 울트라를 사쎌요.

2. Context Disambiguation

- 3:10 [세 시 십 분] (Time: 3:10 PM) ↔ [삼 대 십] (Match Score 3:10)
- 3번 [삼번] (Identifier: Exit No.3) ↔ [세번] (Counter: 3 times)
- 2호선 [이호선] (Route: Line 2) ↔ [두 개] (Quantity: 2 units)

OOV Syllabification & Product Readout

- **ABCTechnology**: Automatic subword syllabification for unlisted English
[에이비씨 테크놀로지]
- **Galaxy Z fold8 ultra**: Natural Korean reading for IT model numbers [갤럭시 지 폴드에인 울트라]

3. Speed & Resource Footprint: 1.86 ms (avg. 32 chars) and 9.67 ms (avg. 618 chars) with ~130 MB memory on a single CPU thread.

Slide 03. Why SNAP Frontend Matters

Overcoming text preprocessing (TN/G2P) bottlenecks and limitations of existing approaches behind modern acoustic models

1. Acoustic Model Advancement vs. Text Preprocessing Bottleneck

Recent breakthroughs in acoustic architectures and neural vocoders have dramatically improved raw audio quality.

However, without front-end preprocessing that reliably determines "**what exact phonetic sequence to articulate (G2P/TN)**", synthesis exhibits **crystal-clear audio delivering incorrect or unnatural Korean pronunciations**.

Rule/Regex-based Systems

While fast and deterministic for fixed rules, regex fails on context-dependent **heteronyms (대까/대가, 신포/신고)**, compound alphanumeric formats (10:12), and loanword phonotactics.

Global Open Source (eSpeak-ng)

Standard across Piper, Coqui, and Kokoro, eSpeak-ng fails to capture **agglutinative grammar and complex Korean phonological alternations**, yielding robotic and broken speech.

Large Language Models (LLM)

Prompting LLMs to expand text into Hangul phonemes introduces **prohibitive latency unsuitable for real-time streaming**, while still occasionally misinterpreting syntactic boundaries.

3. The SNAP Solution: Ultra-low Latency Real-time Hybrid Architecture

Synergy of a **lightweight miniBERT backbone** for semantic context extraction and a **C++ native rule engine** encoding all 30 Articles of Standard Korean Pronunciation.

Guarantees contextually precise phonemic streams within **1 to 3 milliseconds**, empowering real-time conversational AI and mission-critical voice services.

Slide 04. SNAP-Powered Application Ecosystem

From commercial cloud TTS correction to on-device neural synthesis, STT data generation, and dubbing pipelines

1. SNAP-Edge-TTS : Cloud Voice Disambiguation

Pipeline Integration with High-End Commercial TTS

LIVE DEMO: https://huggingface.co/spaces/softguy777/snap_voice_demo4

Commercial cloud TTS engines frequently misread heteronyms and numeric formats in complex contexts. Integrating SNAP upstream resolves contextual ambiguity, elevating end-to-end service reliability.

2. SNAP-Piper : Lightweight On-Device Synthesis

Replacing eSpeak-ng with SNAP & Retraining Piper

LIVE DEMO: https://huggingface.co/spaces/softguy777/snap_voice_demo2

Replaces the flawed default eSpeak-ng frontend in Piper. Feeding richer linguistic features (POS tags, prosodic phrasing, vowel length) yields vastly superior naturalness while retaining compact model size.

3. SNAP-STT / ITN : Data Synthesis for ASR Pre/Post-processing

Large-Scale Automated Text-to-Phoneme Alignment Generation

Automatically synthesizes massive pairs of "surface text ↔ normalized pronunciation" from raw corpora. Enables compact neural models to achieve state-of-the-art ASR transcription and Inverse Text Normalization (ITN).

4. SNAP-Dubber : Multi-Speaker Video Dubbing Pipeline

Colloquial Spoken Conversion for Translated Scripts

Converts translated foreign scripts from stiff literary written style into fluid Korean conversational pronunciations. Synchronizes speaker voice cloning with video timecodes for end-to-end multi-speaker dubbing.

Slide 05. Standard Phonological Rules & Precision Conversion

100% compliance with all 30 Articles of National Institute of Korean Language (NIKL) + Contextual Heteronyms

Vowel Length (Chapter 3)

- 밤 [밤:] [time: night] ↔ 밤 [밤] [fruit: chestnut]
- 눈 [눈:] [weather: snow] ↔ 눈 [눈] [body: eye]

Coda Neutralization (Ch. 4)

- 맛없다 [마덜따] ↔ 옷이 [오시]
- Tensification in Verb Stems (Ch. 6):
신발을 신고 [신꼬] ↔ 안기다 [안기다], 갈등 [갈
똥]

Assimilation & Sai-siot (Ch. 5/7)

- Assimilation (Ch. 5): 국물 [궁물], 신라 [실라] ↔ 생
산량 [생산량]
- Sai-siot & 'ㄴ'-Addition (Ch. 7):
나뭇잎 [나문닙], 숨이불 [숨니불] ↔ 냇가 [내까]

2. Context Disambiguation for Heteronyms

[Part-of-Speech Bifurcation]

- 신발을 신고 [신꼬] (Verb: putting on shoes) ↔ 경찰에 신고 [신고]
(Noun: report to police)

[Semantic Meaning Bifurcation]

- 희생의 대가 [대까] (Cost/Price) ↔ 서예의 대가 [대가] (Master Artist)
- 잠자리에 들다 [잠짜리] (Bed/Sleep) ↔ 고추잠자리 [잠자리] (Dragonfly)

3. Loanwords & Popular Colloquial Readings

[Popular Loanword Tensification]

- 버스 → [빠스], 서비스 → [써비스], 카페 → [까페]

[Unregistered Tech Compound Syllabification]

- FastAPI → [패스트에이피아이]
- PyTorch → [파이토치]

Slide 06. Text Normalization (Numbers, Units & Symbols)

Comprehensive engine for 100+ unit systems, colloquial abbreviations, formatted digit expansion, and particle selection

1. Number Readings & Complex IT Model Names

- **Tech & Appliance Names:** iPhone 16 Pro Max [십육], Galaxy Z fold8 [지 폴드에이트], PS5 [피에스파이버]
- **Serial Numbers & Versions:** RTX 4090 [사공구공] (sequential), v2.4.1 [버전 이점 사점 일]
- **Formatted Digits:** Phone 010-1234-5678 [공일공...], IP 192.168.0.1 [일구이점...]

2. Context-Sensitive Numbers & Symbols

- **Sino vs. Native Numerals:** 3번 버스 [삼번] (No.3 Bus) ↔ 3번 반복 [세번] (3 repeats), 2호선 [이호선] ↔ 2대 [두대]
- **Punctuation Semantics:** 10:12 [열 시 십이 분] (10:12 AM) ↔ [십 대 십이] (Score 10:12), 1/2 [이분의 일] ↔ 3/4분기 [삼사분기]
- **Hyphen Bifurcation:** 010-1234-5678 (phone) ↔ -10°C [영하 십도] (negative) ↔ 10-20개 [십에서 이십개] (range)

3. 100+ Unit Systems & Colloquial Styling

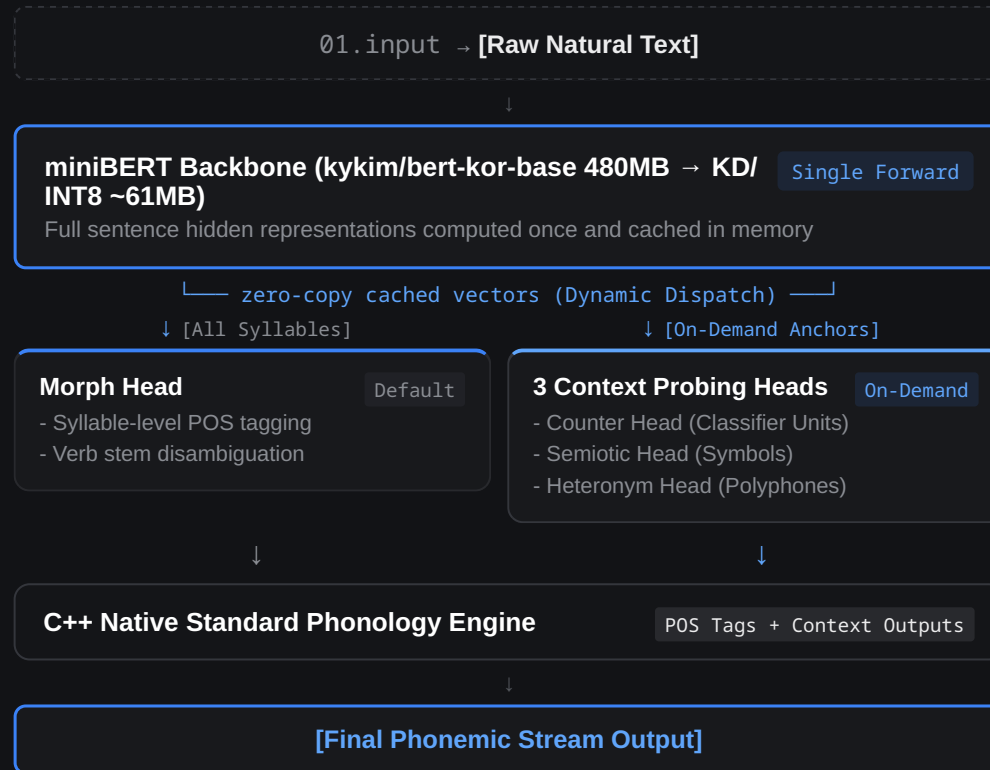
- **6 Broad Unit Domains:** Length/Area, Mass/Density, IT/Data, Time, Power, Percentages
- **Standard ↔ Colloquial Abbreviation:**
120km/h [백이십 킬로미터] ↔ [백이십 키로]
100% [백 퍼센트] ↔ [백 프로], 16GB [십육 기가]

4. Currency & Postposition (Josa) Auto-Correction

- **Currency Symbols & Multipliers:**
\$2.5M → 이쯤오 밀리언 달러, ₩100K → 백 케이 원
- **Phonologically Aligned Postpositions:**
\$2.5M은 → 달러는 (remedies naive regex error '달러은')
15km는 → 킬로미터는, 15km/h로 → 시속 십오 킬로미터로

Slide 07-A. miniBERT Backbone & Probing Head Pipeline (1/2)

Hybrid architecture executing full sentence hidden state computation once, dispatching probing heads only on anchor tokens



2. Model Compression & On-Demand Inference

• Backbone Compression:

kykim/bert-kor-base (480MB) → Knowledge Distillation (KD)
+ INT8 Quantization → **~61MB miniBERT (61M parameters)**

• Morph Head (Morphology Tagging):

Syllable-level BIO-POS sequence labeling confines pronunciation-altering **verb classification errors to only 0.48%**

• Anchor-Triggered On-Demand Execution:

After a single backbone pass, specialized heads evaluate exclusively at detected anchor positions (numbers, symbols, heteronyms), eliminating computational redundancy.

Slide 07-B. miniBERT Backbone & Probing Heads (2/2: Accuracy)

Measured accuracy against NIKL Gold Standards across 4 Probing Heads and in-depth Morph Head error breakdown

3. Measured Accuracy by Probing Head (NIKL Gold Standard)

PROBING HEAD	DOMAIN TARGET	ACCURACY	KEY ADVANTAGE OVER RULE/LATTICE
Counter Head	Numeral-Classifier Units (Sino/Native)	99.11%	8 primary units & 50+ numeral context bifurcations
Semiotic Head	Special Symbols (:, /, -, ~)	98.21%	Disambiguates Time/Score, Fraction/Unit, Phone numbers
Heteronym Head	Homograph Tensification Decisions	97.25%	Semantic discrimination: 대가, 시가, 잡자리
Morph Head	Syllable Detailed Part-of-Speech (BIO-POS)	93.98%	Pronunciation-critical verb errors kept under 0.48%

Morph Head Error (6.02%) Breakdown & Verification

The overwhelming majority of errors (4.5%p) correspond to benign noun-class confusions (NNG ↔ NNP) which have **zero impact on pronunciation**. **Actual verb stem errors influencing phonological alternations remain at only 0.48%**, ensuring robust, stable acoustic model input generation.

Slide 08. Performance Benchmarks & Hardware Efficiency

Single CPU thread verified millisecond-level latency and zero GPU VRAM consumption

1. Environment & Resource Footprint (Intel Core i7-13700K, Single CPU Thread)

- **Total Model File Size: ~62 MB** (miniBERT Backbone + 4 Probing Heads combined)
- **Resident Memory (Working Set): ~130 MB** (ONNX Runtime + C++ Lexicons fully loaded)
- **Zero GPU Contention: CPU Only (0 MB GPU VRAM)** — 100% GPU resources dedicated to acoustic synthesis models

2. Latency & Throughput Benchmark by Sentence Length (Continuous Single CPU Thread)

LENGTH CATEGORY	AVG. CHARACTERS	AVG. LATENCY	THROUGHPUT	TARGET APPLICATION
Short (≤ 10 words)	32.1 chars (10~65)	1.86 ms	538.8 FPS	Conversational AI / Voicebots
Standard (≤ 20 words)	64.8 chars (38~109)	2.57 ms	389.0 FPS	Voice Guidance & News Bulletins
Medium (≤ 30 words)	104.4 chars (76~156)	3.32 ms	301.1 FPS	Explanatory & Descriptive Content
Long (≤ 50 words)	153.9 chars (112~199)	4.33 ms	230.7 FPS	Documentary Narration
Extended (16 sentences)	618.0 chars	9.67 ms	103.4 FPS	Audiobooks & Longform Reading

Slide 09. Domain-Customized Option Controls

Fine-grained control over pronunciation style, speech registers, unit expansions, and prosodic punctuation

1. Pronunciation Style (pronunciation_style)

- **Strict Standard:** 효과 [효과], 버스 [버스], 우리의 희망 [우리의 희망]
- **Modern Standard (Default):** 효과 [효과], 버스 [빠스], 우리의 희망 [우리애 희망]
- **Colloquial:** 시계 [시게], 전화 [저놔], 일하다 [이라다]
- **Vowel Length:** vowel_length: true (밤: ↔ 밤) with SSML prosody integration

2. Unit Expansion Modes (unit_style)

- **standard (Default):** 70kg → 칠십 킬로그램, 100% → 백 퍼센트, 16GB → 십육 기가바이트
- **full:** 120km/h → 백이십 킬로미터 퍼 아워
- **short (Spoken):** 70kg → 칠십 키로, 100% → 백 프로, 16GB → 십육 기가

3. Prosodic Punctuation (prosody_control)

- **Universal Punctuation Correction:** Automatically injects periods (.) and breathing commas (,) to unpunctuated conversational text for natural pauses in non-SSML engines.
- **Precise SSML Tags:** Outputs 3-tier syntactic breaks:
<break strength="weak|medium|strong"/>

4. Conversational Honorific Registers (tone_style)

- **Context-Aware Verb Ending Transformation:**
 - **Polite Spoken (Haeyo-che):** 도착했다 → 도착했어요
 - **Informal (Ban-mal):** 도착했다 → 도착했어
 - **Formal Deferential (Hasipsio-che):** 도착했다 → 도착했습니다

Slide 10. Acoustic Model Input Specifications

Providing 4 standardized phonetic and prosodic output formats tailored for neural acoustic synthesizers

1. Hangul Syllable Phonemes (phoneme)

Normalized surface Hangul with full phonological alternations applied.

Input: "2호선 3번 출구로 오세요"
→ "이호선 삼번 출구로 오세요."

2. Decomposed Jamo Sequence (jamo)

Fully decomposed onset-nucleus-coda sequence for character/jamo-level models.

Input: "2026년"
→ "ㅇ | ㄷ | ㄴ | ㅇ | ㅅ | ㅁ | ㄴ | ㅇ | ㄷ | ㄴ"

3. International Phonetic Alphabet (ipa)

Standardized IPA phonetic symbols for multilingual acoustic foundations.

Input: "2026년"
→ "[it͈ʰʌnisimɲunɲʌn]"

4. Prosodic Phrasing & SSML (prosody)

Dual mode: universal breathing punctuation or syntactic W3C SSML tags.

<peak>정부는 <break strength="weak"/> 국무회의를 열고
<break strength="medium"/> 확정했습니다.</peak>

Slide 11. Serving Architecture & Integration Modalities

From containerized microservices to in-process C++ embedded SDKs across all deployment targets

1. Delivery Modalities & Dynamic Custom Dictionary

- **Cloud Microservices (HTTP REST API / Docker Container):**
 - Public testbed API server active in production
 - Private dedicated cloud / on-premise container cluster deployments
- **C/C++ Native Shared Libraries (In-Process Embedded):**
 - Embedded SDK (.lib, .dll, .so) directly linked to local processes
- **Runtime Dynamic Custom Dictionary (custom_dict):**
 - Immediate zero-downtime override for proprietary brand names, acronyms, and proper nouns without service restarts.

2. High-Performance Container Serving Optimizations

- **Non-Blocking Asynchronous I/O Pipeline:**
 - Decouples network socket event loops from deep learning inference worker pools, preventing I/O queue stalls.
- **Core-Pinned Fixed Worker Pool:**
 - Pre-allocates workers strictly matched to physical CPU cores, preventing thread explosion and context-switching overhead.
 - Dedicated core execution prevents L1/L2/L3 cache invalidation, maximizing memory locality.

Slide 12. Conclusion & Summary

Establishing the standard for Korean text preprocessing through standard phonology rules and deep learning context awareness

1. SNAP v2.0 Executive Summary

- **Uncompromising Standard Compliance & Colloquial Coverage:**

Holistically supports rigorous phonological rules (liaison, tensification, nasalization) alongside spoken conventions, loanwords, and complex alphanumeric units.

- **High Accuracy & Production-Grade Real-Time Throughput:**

Achieves **98%+** accuracy across NIKL evaluation benchmarks with single-threaded latency of **1.86 ms** (>530 sentences/sec).

2. Live Demonstrations & Repositories

- **Real-Time TTS Live Interactive Demo:**

https://huggingface.co/spaces/softguy777/snap_voice_demo4

- Enter arbitrary natural text in browser and listen to synthesized speech immediately.

- **Public Testing REST API Server:**

- Active endpoint operational for live validation (POST /v1/normalize)

- **Official GitHub Repository:**

<https://github.com/snap-libs/snap>

- C++ build guides, Python bindings, benchmark datasets, and technical documentation.

Appendix 01. Context Disambiguation Benchmarks

Focused benchmark metrics across the highest-frequency ambiguous linguistic groups in Korean daily and business text

1. Heteronyms (Accuracy: 97.25%)

Identical spelling (NNG) resolving to plain vs. tensified sounds by syntactic context:

- **대가:** 희생의 대가를 치르다 [대까] (Price/Cost) ↔ 서예의 대가를 만나다 [대가] (Master)
- **시가:** 부동산 시가 총액 [시까] (Market Price) ↔ 조선 시대 시가 문학 [시가] (Poetry)
- **성적:** 성적 수치심 [성쩍] (Sexual) ↔ 시험 성적 [성적] (Grades)
- **잠자리:** 잠자리에 들다 [잠짜리] (Bed) ↔ 고추잠자리 [잠자리] (Dragonfly)
- **Other Targets:** 열병([열뽕]/[열병]), 송장([송짱]/[송장]), 지적([지쩍]/[지적]), 감기([감끼]/[감기]), 안다([안따]/[안다])

2. Numeral Classifiers (Accuracy: 99.11%)

Bifurcates digits into Sino-Korean or Native Korean counting words based on following unit noun:

- **번:** 3번 버스 [삼번] (No.3 Bus) ↔ 3번 반복 [세번] (3 times)
- **대:** 20대 청년 [이십대] (20s Age) ↔ 차량 2대 [두대] (2 vehicles)
- **동:** 101동 [백일동] (Bldg 101) ↔ 비닐하우스 2동 [두동] (2 greenhouses)
- **장:** 제3장 [제삼장] (Chapter 3) ↔ 종이 3장 [세장] (3 sheets)
- **Other Units:** 점([사뽀오점]/[세점]), 단([사단]/[두단]), 기([제오기]/[두기]), 세트([세트 일]/[두세트])

3. Punctuation & Symbols (Accuracy: 98.21%)

Decodes identical symbols into natural colloquial words by analyzing neighboring tokens:

- **Colon (:):** 10:12 [열 시 십이 분] (Time) ↔ 10:12 [십 대 십이] (Score)
- **Slash (/):** 3/4분기 [삼사분기] (Q3) ↔ 1/2 [이분의 일] (Half) ↔ km/h [킬로미터 퍼 아워]
- **Tilde (~):** 10~20m [십에서 이십미터] (Range)
- **Hyphen (-):** 02-1234-5678 [공이 일이삼사...] (Phone) ↔ -10°C [영하 십도] (Minus)

Appendix 02. Phonological Standards & Colloquial Rules

Boundary-aware liaison, coda neutralizations, and selective support for prescriptive standard vs. everyday colloquial forms

1. Morphological Boundary & POS-Aware Phonology

Systematic application of linguistic rules beyond surface character replacement:

- **Substantive Morpheme Resyllabification (Article 15):**

겉옷 → [거돋], 맛없다 → [마덜따] (*Substantive boundary: neutralized then liaised*)

옷이 → [오시], 꽃을 → [꼬출] (*Particle/Ending: directly liaised without neutralization*)

- **Sino-Korean 'ㄹ' Tensification (Article 26):**

갈등 → [갈똥], 결정 → [결똥] (*Sino-Korean tensification*)

살다 → [살다], 발달 → [발달] (*Native verbs & standardized exceptions maintain plain sound*)

- **Verb Stem Ending Tensification (Article 24):**

신발을 신고 → [신꼬] (*Stem coda triggers tensification in suffixes*)

안기다, 감기다 → [안기다], [감기다] (*Passive/Causative affixes remain plain*)

- **Adnominal Ending '-(으)ㄹ' Tensification (Article 27):**

할 것을 → [할 꺼슬], 갈 데가 → [갈 떤가] (*Tensifies following dependent nouns*)

2. Prescriptive Standards vs. Colloquial Acceptance

Customizable balance between formal correctness and natural everyday speech:

- **Standard Vowel Length Distinction (Ch. 3, Optional):**

밤 [밤:] [night] ↔ 밤 [밤] [chestnut], 눈 [눈:] [snow] ↔ 눈 [눈] [eye]

- **Dual Standard Pronunciations Supported:**

효과 [효과/효파], 교과서 [교과서/교파서], 우리의 [우리의/우리예]

- **Popular Loanword Tensification (Colloquial Mode):**

버스 → [뻐스], 서비스 → [써비스], 카페 → [까페]

- **Consonant Weakening & Assimilation (Colloquial Mode):**

전화 → [저놔], 일하다 → [이라다], 신문 → [심문]

Appendix 03. Loanwords, Model Names & Formats

Out-of-vocabulary (OOV) phonetic recovery, consumer tech model readout, particle selection, and symbol cleanup

1. Unlisted English Words (OOV) Automatic Syllabification

- **CamelCase & Subword Decomposition:**

ABCTechnology → ABC + Technology → [에이비씨 테크놀로지]

- **Consonant-Vowel (CV) Pattern Recovery:**

FastAPI → [패스트에이피아이], PyTorch → [파이토치]

2. Established Readout for Product Model Numbers

- **Common Reading of Mixed Alphanumeric Tech Terms:**

iPhone 16 Pro Max → 아이폰 십육 프로 맥스 → [아이폰 십육 프로 맥쓰]

Galaxy Z fold8 ultra → 갤럭시 지 폴드에이트 울트라 → [갤럭시 지 폴드에일 울트라]

RTX 4090 → [알티엑스 사공구공], v2.4.1 → [버전 이점 사점 일]

3. Particle (Josa) Concordance on Number/Unit Compounds

Dynamically corrects Korean case particles according to the final coda consonant of expanded Hangul text:

- \$2.5M은 → 이점오 밀리언 달러 + 는 → "달러는" (*remedies naive rule error* '달러은')

- 15km는 → 십오 킬로미터는, 15km/h로 → 시속 십오 킬로미터로

4. Non-Phonemic Symbols & Parenthetical Sanitization

- **Parenthetical Glosses:** 대한민국(Republic of Korea) → 대한민국

(Removes redundant annotations)

- **Decorative Symbol Scrubbing:** Bullets (•, ►), emojis, and unpronounceable formatting glyphs safely sanitized.

Appendix 04-A. Standard Korean Pronunciation Rules (Part 1: Ch. 1 ~ 4)

Complete technical specification covering General Principles to Coda Pronunciations (Articles 1 to 16)

CHAPTER	ARTICLES	KEY PROVISIONS & PHONOLOGICAL RULES	VERIFICATION LEXICON	SNAP IMPLEMENTATION
Chapter 1: General	Art. 1	Fundamental principles of standard pronunciation	Standard Seoul modern usage baseline	100% baseline compliance
Chapter 2: Consonants & Vowels	Art. 2~5	19 consonants, 10 monophthongs & diphthongs Non-initial '의' [이], genitive particle '의' [에]	주의 [주이], 우리의 [우리예] 시계 [시게], 희망 [히망]	POS context tagging + colloquial options
Chapter 3: Vowel Length	Art. 6~7	Semantic vowel length & compound shortening	밤 [밤:] [night] ↔ 밤 [밤] [chestnut] 눈 [눈:] [snow] ↔ 눈 [눈] [eye]	Built-in dictionary + SSML duration tags
Chapter 4: Coda Pronunciation	Art. 8~16	7 representative coda stops, consonant cluster simplification, 'ㅎ' aspiration, substantive boundary liaison	꽃 [꼰], 닭 [닥], 좋다 [조타] 겉옷 [거돋], 맛없다 [마덜따]	Zero-allocation C++ stack liaison & Art. 15 boundary discrimination

Appendix 04-B. Standard Korean Pronunciation Rules (Part 2: Ch. 5 ~ 7)

Complete technical specification covering Sound Assimilation to Sound Addition (Articles 17 to 30)

CHAPTER	ARTICLES	KEY PROVISIONS & PHONOLOGICAL RULES	VERIFICATION LEXICON	SNAP IMPLEMENTATION
Chapter 5: Assimilation	Art. 17~22	Palatalization, nasalization, liquidization Sino-Korean liquidization exceptions (Art. 20)	굳이 [구지], 국물 [궁물] 신라 [실라] ↔ 생산량 [생산냥]	Full phonetic assimilation engine + Sino-Korean suffix inhibition
Chapter 6: Tensification	Art. 23~28	Post-stop tensification, verb stem tensification, Sino-Korean 'ㄹ' tensification, adnominal '-(으)ㄹ'	국밥 [국빔], 신고 [신키] 갈등 [갈똥], 할 것을 [할 꺼슬]	Neural BIO-POS sequence tagging for accurate stem/ending classification
Chapter 7: Sound Addition	Art. 29~30	Compound 'ㄴ' addition Sai-siot orthography pronunciation	숨이불 [숨니불], 나뭇잎 [나문닙] 넷가 [내까], 숫자 [수짜]	Automatic morpheme boundary detection & standard sai-siot normalization

Appendix 05-A. Morph Head Error Decomposition (1/2: Quant & Ambiguity)

Quantitative error decomposition on NIKL 128k syllable dataset and intrinsic linguistic ambiguity analysis

1. Quantitative Error Breakdown (Accuracy: 93.98%, Total Error: 6.02%)			
Error Category	Portion of Text	Tag Pair & Error Pattern	Phonetic (G2P) Impact
Noun-class Confusions	4.52%p (75.1% of errors)	Common ↔ Proper noun (NNG ↔ NNP) Common noun ↔ Root (NNG ↔ XR)	Negligible (0% impact) Identical liaison and coda rules
Symbol Tagging Variations	1.02%p (16.9% of errors)	Punctuation (SF/SP) ↔ Symbol (SW/SO)	Zero impact (0%) Absorbed during text normalization
Pronunciation-critical Verbs	0.48%p (8.0% of errors)	Verb stem (VV/VA) ↔ Substantive Noun	Direct phonetic effect Tension rule (Art. 24) sensitivity

2. Intrinsic Linguistic Ambiguity in Korean Morphology

While statistically critical verb errors stand at only 0.48%, natural language harbors complex latent errors:

- **Local Ambiguity Coexistence:** Examining only local syllable windows frequently yields multiple syntactically valid morpheme candidates.
- **Global Inconsistency:** A parse valid in an isolated subphrase can distort the final phonetic stream when evaluated against the whole sentence predicate structure.

Appendix 05-B. Morph Head Error Decomposition (2/2: Safety Net)

C++ deterministic rule post-processing safety nets and empirical case defense verification ('신고')

3. SNAP Hybrid Post-Processing Safety Net

SNAP avoids sole reliance on neural probabilities.

Immediately following neural tagging, candidates pass through a **C++ Trie Lexicon verification** and an **orthographic exception table**. Deterministic rule post-processing corrects local ambiguities within the full sentence context to finalize the phonetic stream.

4. Real-World Case Defense: The "신고" (Shinko vs. Shingo) Ambiguity

• Failure Modes in Standard Parsers:

- Misclassifying "신발을 신고 나갔다" (wearing shoes) as a noun yields [신고] (omitting tensification; correct: [신표]).
- Conversely, misparsing "신고식" (initiation ceremony) as a verb yields [신표식] (hyper-tensification; correct: [신고식]).

• SNAP Dual-Layer Defense:

miniBERT Bidirectional Context (Layer 1) + **Verb Lexicon & Suffix Filter (Layer 2)** reliably separates [신표] and [신고식] without error.